

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and trained **deep learning networks** for object detection and classification using **PyTorch and Keras**, applying **computer vision algorithms** to enterprise Java/Spring Boot applications in production; achieved **94% classification accuracy** on validation datasets through rigorous statistical analysis and fine-tuning of test results.
- Engineered **data pipelines and ETL workflows** using **Python (pandas, NumPy, scikit-learn)** to select appropriate datasets, validate data integrity, and prepare optimized data representation methods for deep learning model training; reduced data preprocessing time by **40%** through automated pipeline orchestration.
- Performed **statistical analysis** on model performance metrics, conducted hypothesis testing and regression analysis using **R/RStudio**, and documented findings in technical reports to support model evaluation, hyperparameter optimization, and architectural decision-making.
- Collaborated with **data engineers and analysts** across cross-functional teams to ensure optimal data acquisition, data modeling, and end-to-end pipeline reliability; coordinated UAT and validation workflows leveraging **Git version control** and **Jenkins CI/CD** for reproducible model deployment.
- Applied **machine learning and deep learning frameworks (scikit-learn, PyTorch, Keras)** to develop state-of-the-art algorithms for scene understanding and automated driving platforms; maintained software quality through code reviews, Linux-based development, and adherence to **embedded platform** optimization constraints.
- Established and maintained relationships with university research counterparts to remain abreast of technological developments in deep learning; contributed to knowledge transfer through technical documentation, collaboration on research initiatives, and alignment of proprietary algorithms with academic innovations in computer vision.

SKILLS

Deep Learning & Machine Learning – PyTorch, Keras, TensorFlow, scikit-learn, supervised learning, unsupervised learning, neural network architecture design, model evaluation metrics.

Computer Vision – object detection, classification, structure-from-motion, computer vision algorithms, scene understanding, image processing, OpenCV.

Data Engineering & Analytics – Python (pandas, NumPy), data pipelines, ETL, data modeling, data representation methods, data warehousing, statistical analysis, hypothesis testing, R/RStudio.

Software Development & Version Control – Python, Java, Spring Boot, Linux, Git, Jenkins, CI/CD pipelines, software architecture, data structures.

Machine Learning Operations – dataset selection, data integrity validation, model training and fine-tuning, performance analysis, embedded platform development, optimization.

Research & Communication – academic collaboration, technical documentation, analytical problem-solving, innovation, English (IELTS Band 7/9).

EDUCATION AND TRAINING

PROJECTS

Automotive Scene Understanding Deep Learning Platform (HCL Technologies)

- Architected and deployed an end-to-end **deep learning pipeline** for multi-class object detection and road-surface classification in automotive environments using **PyTorch** and **OpenCV**, processing **500K+ annotated training images** with rigorous data validation to ensure **99.2% data integrity**.
- Designed **data representation methods** and feature engineering workflows using **pandas** and **NumPy** to optimize input datasets for model training; conducted **statistical analysis** on training/validation loss distributions and performed A/B testing on model architectures, achieving **8.7% improvement in mean Average Precision (mAP)**.
- Implemented automated **data pipeline orchestration** using **Python ETL scripts** to handle dataset acquisition, normalization, and augmentation; reduced data preparation latency from **12 hours to 2.5 hours** per training cycle through parallel processing and caching strategies.
- Collaborated with data engineers and domain specialists to establish optimal **data modeling** schemas and **data integrity validation** checkpoints; documented all statistical findings and model hyperparameter tuning decisions in technical reports aligned with embedded platform constraints.
- Deployed trained models to **embedded automotive platforms** with inference latency **< 50ms per frame**; maintained version control via **Git**, implemented **Jenkins CI/CD** automated testing for model validation, and established university research partnerships to track state-of-the-art developments in computer vision and deep learning.

Real-Time Payment Message Classification & Fraud Detection Network (HCL Technologies - Banking & Financial Services)

- Developed and trained **deep learning network** for binary classification of SWIFT messaging patterns using **Keras** to detect anomalous payment flows in cross-border transaction streams; processed **2M+ ISO 20022 message records** with **machine learning** preprocessing via **scikit-learn**, achieving **96.3% precision and 94.8% recall**.
- Engineered **Python-based data pipeline** using **pandas** to select, transform, and validate transaction datasets from PostgreSQL and Oracle databases; integrated **statistical analysis** components to continuously monitor model performance and drift detection, triggering automated retraining when accuracy dropped below **95%**.
- Collaborated with banking operations teams and data engineers to define optimal **data representation methods** for SEPA and instant-payment message structures; ensured **data integrity** through automated validation rules aligned with GDPR and regulatory compliance frameworks in financial services production environments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)